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Target regions for silvoarable agroforestry in Europe

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Holm oak (*Quercus ilex*)

ABSTRACT

Silvoarable agroforestry (SAF) has recently been proposed as an alternative land-use system for Europe. Data on soil, climate, topography, and land cover were integrated in a geographic information system (GIS) to identify agroforestry target regions where (i) productive growth of trees (*Juglans* spp., *Prunus avium*, *Populus* spp., *Pinus pinea*, and *Quercus ilex*) in SAF systems could be expected and where (ii) SAF systems could potentially reduce the risk of soil erosion, nitrate leaching and increase landscape diversity.

The analysis shows that the investigated tree species could grow productively in SAF systems on 56% of the arable land throughout Europe (potential productive tree growth area). 80% of the European arable land was classified as potential risk area for soil erosion, nitrate leaching, and/or landscape diversity. Overlaying potential productive tree growth areas with the arable land that was considered as environmental risk area yielded target regions. They were found to make up about 40% of the European arable land and thus SAF could contribute to soil protection on 4%, to mitigate nitrate leaching on 18% and to increase landscape diversity on 32% of European arable land.

Although limited by constrained data availability, the study shows that SAF could be implemented in a productive way throughout Europe and that it could help resolve some of the major land-use problems. The environmental benefits could justify the support of SAF by subsidies.

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1. Introduction

In Europe, agro-political discussions have been conducted for many years about overproduction, subsidies, and resource protection and the future of farmers in general. The importance of agroforestry in this context is still marginal despite its great problem-solving potential to help farmers in certain regions in Europe (Le Houérou, 1990; Dupraz, 1994; Étienne, 1996).

Agroforestry in general is defined as a land-use system in which two or more plant species interact, one of the species being perennial and having a woody consistency (Somarriba,

1992). Silvoarable agroforestry (SAF) comprises agricultural systems where trees and crops are cultivated on the same land area. Successful agroforestry systems produce high-value products (e.g. good-quality timber, food, or nuts) (Lundgren, 1982; Lundgren and Raintree, 1982; Carruthers, 1990; Nair, 1993; Sinclair, 1999).

Traditional agroforestry systems are still quite widespread in Europe (Byington, 1990; Dupraz and Newman, 1997). One important traditional agrosilvopastoral system is the Dehesas in Spain, where oak trees are combined with cereals and livestock. Traditional systems, for instance “Streubobst” orchards, are also found in temperate Europe (Herzog, 1998). In con-

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trast to traditional systems, modern agroforestry systems are hardly adopted by farmers. One reason for this is that agroforestry is not supported by subsidies, whereas agriculture and forestry receive government support in all countries. To decide whether and where to promote agroforestry, policy makers and farmers have to know where silvoarable agroforestry is possible and where it makes sense in terms of reducing environmental problems such as soil erosion or nitrate leaching.

The potential advantages of agroforestry in temperate and Mediterranean climatic zones are multifaceted. Agroforestry diversifies the agricultural trade and market and reduces overproduction of agricultural commodities. It could increase landscape diversity and enhance biodiversity in certain landscapes where the lack of diversity is quite evident. Other resource protection issues include, for example, the reduction of soil erosion and nitrate leaching in intensively managed agricultural landscapes (Anderson and Sinclair, 1993; Nair, 1993; Gordon and Newman, 1997; Herzog, 1997). Agroforestry can reduce nitrate leaching through the potential of the tree roots to recover nitrogen from below the crop rooting zone, and increased water uptake from the system (Sanchez, 1995; van Noordwijk et al., 1996; Rowe et al., 2001). Furthermore, agroforestry can contribute to enabling farmers to cultivate poor soils as arable land and to protect large areas from afforestation, through the soil improving effects of trees (e.g. enhancing soil organic matter, improving water holding capacity, and increasing nutrient inputs through nitrogen fixation) (Young, 1997). This can make marginal areas economically more attractive and can help preserve appealing landscapes for recreation. Numerous studies support the environmental value of agroforestry within the European context (e.g. Telleria and Santos, 1995; Burgess, 1999; Herzog, 2000; Pinto-Correia, 2000; Shakesby et al., 2002; Palma et al., 2007).

There are two approaches to map potential distribution areas for plants: either directly from databases of actual observations which are then generalised, or indirectly by mapping areas where ecological growth conditions are fulfilled. The first approach was implemented by, e.g., Hijmans and Spooner (2001), who analysed the distribution of wild potatoes using geographic information system (GIS) with a large database of locations where wild potatoes were observed. The second approach is usually limited to climatic criteria; soil – the other main ecological factor – is not included due to the lack of appropriate and harmonised soil maps, especially for larger regions. Booth et al. (1989) and Booth and Jones (1998) defined climatically suitable areas for particular tree species in Africa and Latin America. Based on a description of the trees' requirements of climatic conditions they generated maps showing locations which are suitable for these tree species. Booth et al. (2002) and Booth and Jovanovic (2002) proposed a world-mapping program (WORLD), which indicates locations satisfying up to six climatic criteria important for tree species selection. One of their studies is limited to the African continent. The other study was undertaken at a global scale and here the European countries are included at a very coarse level. Ellis et al. (2000) used GIS systems to model agroforestry opportunities and potential tree species in Florida.

The objective of this paper is to model target regions for SAF systems in Europe, based on environmental analysis. We tried to answer two questions:

- (1) Where is it possible to implement agroforestry systems and expect the productive growth of walnut (*Juglans* spp.), wild cherry (*Prunus avium*), poplar (*Populus* spp.), Italian stone pine (*Pinus pinea*), and holm oak (*Quercus ilex*)?
- (2) Where in Europe can new SAF systems mitigate negative environmental effects of agricultural land use on nature and landscape?

The potential agroforestry areas were identified through a GIS-analysis based on digitally available geo-information about land cover, topography, climate, and soil properties. Due to data availability and quality, this assessment was “coarse-grained” with a resolution of 1100 m × 1100 m. Maps were produced and analysed to identify as well as visualise the potential of silvoarable agroforestry at the European scale.

2. Material and methods

2.1. The general outline

The aim was to define arable landscapes where trees can both grow and have an economic relevant yield. In these areas, an analysis of the environmental problems was undertaken to define the target regions for silvoarable agroforestry.

Target regions were defined overlaying (Fig. 1):

- (i) Arable landscapes.
- (ii) Regions where productive tree growth in an agroforestry setting is possible. This is based on the following natural conditions, where the trees can grow productively:
 - topography;
 - climate;
 - soil.
- (iii) Regions where environmental problems exist which agroforestry can help to solve. This is based on an analysis of environmental problems:
 - high risk for soil erosion;
 - high nitrate concentration in the groundwater;
 - low landscape diversity.

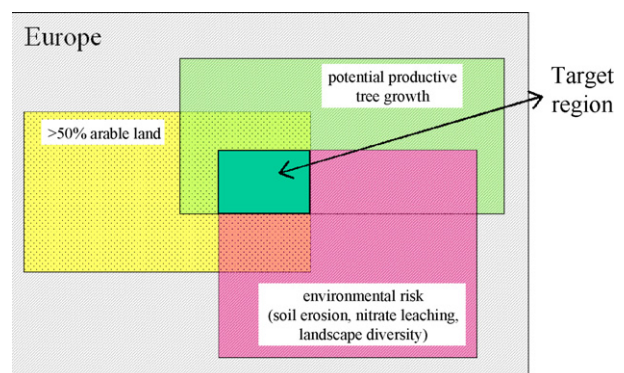


Fig. 1 – Concept to identify target regions for silvoarable agroforestry at the European scale.

Selecting appropriate tree species is one of the most important steps in establishing agroforestry systems. Five tree species, that are known to have potential for use in agroforestry, were investigated. Walnut (*Juglans* spp.), wild cherry (*P. avium*) and poplar (*Populus* spp.) are mostly appropriate for the temperate climate whilst Italian stone pine (*P. pinea*) and holm oak (*Q. ilex*) are typical trees for the Mediterranean region.

2.2. Europe as research area

The research area was defined as the European continent but was constrained by data availability. Consistent datasets could be compiled for the following countries: Albania, Andorra, Austria, Belgium-Luxembourg, Bulgaria, Czech Republic, Denmark, Estonia, Faeroe Island, Finland, France, Germany, Greece, Hungary, Irish Republic, Italy, Latvia, Lithuania, Malta, The Netherlands, Norway, Poland, Portugal, Romania, Russia (small part around Kaliningrad, between Latvia and Poland), Slovakia, Slovenia, Spain, Sweden, Switzerland, and The United Kingdom.

The digital administrative boundaries of the countries were based on the ESRI-dataset (ESRI, 2001). The variables and datasets used are summarised in Table 1.

2.3. Definition of arable land

The analysis of target regions for silvoarable agroforestry in Europe was restricted to arable landscapes as defined in the PELCOM land cover map (Mücher, 2000). Pan-European Land Cover Monitoring (PELCOM) is a European land cover database with a resolution of 1100 m × 1100 m. The map was developed using NOAA-AVHRR satellite data (Mücher, 2000).

The PELCOM map was used for calculating the area of arable land in Europe and for overlaying the environmental risk areas and the potential tree growth areas (Fig. 1). The PELCOM-classes: “Rain fed arable land”, “Irrigated arable land” and “Permanent crops” were defined as arable land.

For the assessment of landscape diversity (Section 2.5.3) a map with a higher resolution than 1 km² was needed. Therefore, the CORINE land cover map (EEA, 2000) with a resolution of 250 m × 250 m was used. On Level 3 of the CORINE map, arable land comprises the two classes: “Non-irrigated arable land” and “Permanently irrigated arable land”.

2.4. Potential productive tree growth areas

The selection of regions where productive tree growth in an agroforestry setting is possible was based on tree growth requirements. Using the Forestry Compendium (CABI, 2003), criteria were defined to describe optimal growth conditions for the five selected tree species. The information was supplemented by expert assessments. The criteria for potential productive tree growth are listed in Table 2. These criteria are more restrictive than values which can be found in the literature for the minimum requirements for the presence of the five tree species under investigation.

2.4.1. Climatic variables

Climatic parameters are essential when defining target regions for silvoarable agroforestry in Europe. Climate characterises the general conditions for land use by defining, for example, the frequency of frost, the water availability, the length of the growing season, and the temperature distribution.

On a European scale, the most comprehensive high-resolution climate datasets available are the CRU TS1.2 (Mitchell et al., 2004) and the CRU CL2.0 (New et al., 2002). Both datasets contain monthly values and have a resolution of 10' × 10'. At 40°N, the east–west distance of 10' corresponds to 14 km; at 60°N it corresponds to 9 km. The north–south distance of 10' is a constant value of 18.5 km. The CRU CL2.0 dataset contains five variables (precipitation, daily mean temperature, diurnal temperature range, vapour pressure, and cloud cover) for the period 1901–2000. From this dataset annual precipitation, mean annual temperature, mean maximum temperature of the hottest month, and mean minimum temperature of the coldest month were derived for the 30-year-period 1971–2000. Information about wind and frost stems from the CRU TS1.2 dataset (30-year-period from 1961 to 1990). The 10' × 10' climate data were subsequently re-sampled to a 1 km² grid by Mächer et al. (2003) using bilinear interpolation.

To define the potential productive tree growth areas, we focused on the most important climatic variables, which are also used in the classification of bio-geographical regions (Metzger et al., 2005), completed by wind speed, which may particularly affect solitary trees:

- precipitation (mm/year);
- mean annual temperature (°C);
- mean maximum temperature of hottest month (August) (°C);
- mean minimum temperature of coldest month (January) (°C);
- wind (m/s);
- frost (days/year).

2.4.2. Soil variables

Soil information is essential to assess the suitability for potential agricultural use. The digital European soil map (ESB, 2002), based upon an internationally accepted classification, was used to derive the following soil characteristics:

- soil texture;
- water regime;
- soil type;
- minimum depth available for root growth (m);
- minimum soil water capacity (1 m top layer) (mm).

These are the main soil characteristics which rule tree growth mainly through water availability and space for rooting, and which at the same time can be derived from the soil database (ESB, 2002).

An MS-Access database was used as the query database system for the ArcGIS platform (ESRI, 2002). The large amount of data in the European soil database (ESB, 2002)

Table 1 – Variables and datasets used to define target regions for silvoarable agroforestry

Variables	Unit	Dataset	Source	Original resolution	Resampled resolution
Research area					
Europe	–	Europe	ESRI (2001)	Vector data	1100 m
Topography					
Altitude range	m above sea level	DEM from HYDRO 1K	USGS (2003)	1000 m	1000 m
Climatic data					
Precipitation	mm/year	CRU CL 2.0; pre	Mitchell et al. (2004)	10'	1000 m (Mücher et al., 2003)
Mean annual temperature	°C	CRU CL 2.0; tmp	Mitchell et al. (2004)	10'	1000 m (Mücher et al., 2003)
Mean maximum temperature of hottest month	°C	CRU CL 2.0; tmp	Mitchell et al. (2004)	10'	1000 m (Mücher et al., 2003)
Mean minimum temperature of coldest month	°C	CRU CL 2.0; tmp	Mitchell et al. (2004)	10'	1000 m (Mücher et al., 2003)
Wind	m/s	CRU TS 1.2; wnd	New et al. (2002)	10'	1000 m
Frost	Days/year	CRU TS 1.2; frs	New et al. (2002)	10'	1000 m
Soil characteristics					
Soil texture	Classified	European soil database	ESB (2002)	Vector data	1100 m
Water regime	Classified	European soil database	ESB (2002)	Vector data	1100 m
Soil type	Classified	European soil database	ESB (2002)	Vector data	1100 m
Minimum depth available for root growth	m	European soil database	ESB (2002)	Vector data	1100 m
Minimum soil water capacity	mm	European soil database	ESB (2002)	Vector data	1100 m
Land cover					
Arable land	Classified	PELCOM CORINE	Mücher (2000) EEA (2000)	1100 m 250 m	1100 m 250 m
Environmental risk					
Soil erosion	t/ha/year	PESERA	Gobin and Govers (2003)	1000 m	1000 m
Nitrate leaching	Classified	Nitrate vulnerable zone	European Commission (2002)	Vector data	1100 m
Landscape diversity	Classified	PELCOM CORINE	Mücher (2000) EEA (2000)	1100 m 250 m	250 m 250 m

Table 2 – Minimum requirements for potential productive tree growth

Criteria	Unit	<i>Juglans</i> spp.	<i>Prunus avium</i>	<i>Populus</i> spp.	<i>Pinus pinea</i>	<i>Quercus ilex</i>
Altitude range	m	0–900	0–1700	0–1000	0–950	200–1200
Precipitation/year	mm/year	450–1500	580–1800	500–3000	300–800	300–1200
Mean annual temperature	°C	8–19	6–14	6–20	10–18	12–21
Mean maximum temperature of August	°C	20–30	<18–30	22–30	27–35	25–37
Mean minimum temperature of January	°C	>–4 to 4	>–8 to 4	>–10 to 11	>–4 to 10	>–3 to 10
Soil texture ^a	Classified	Medium, medium fine, fine, fine	Medium, medium fine, fine	Coarse, medium, medium fine	Coarse, medium	Coarse, medium, medium fine, fine
Water regime ^b	Classified	Dry, semi-wet	Dry, semi-wet	Semi-wet, wet	Dry, semi-wet, wet	Dry, semi-wet
Soil type	Classified	Arenosol, Cambisol, Chernozem, Fluvisol, Greyzem, Luvisol	Arenosol, Cambisol, Chernozem, Fluvisol, Lithosol, Luvisol, Rendzina, Vertisol	All soil types	Arenosol, Cambisol, Lithosol, Rendzina, Regosol	Acrisol, Arenosol, Calcisol, Cambisol, Fluvisol, Leptosol, Luvisol, Regosol, Rendzina, Vertisol
Minimum soil depth available for root growth	m	>1	>1	>1.5	>1	Not relevant
Minimum soil water capacity of the 1 m of soil	mm	150	100	300	100	50
Wind (maximum average wind speed)	m/s	<7	No information	<7	<9	<9
Frost	Days/year	<100	Tolerant	<180	<80	<100

^a Definition of the soil texture classes—coarse: clay <18% and sand >65%; fine: 35% <clay <60%; medium: 18% <clay <35% and sand >15%, or clay <18% and 15% <sand <65%; medium fine: clay <35% and sand <15%.

^b Definition of water regime (dominant annual average soil water regime class of the soil profile): dry: not wet within 80 cm for over 3 months, nor wet within 40 cm for over 1 month; semi-wet: wet within 80 cm for 3–6 months, but not wet within 40 cm for over 1 month; very wet: wet within 40 cm depth for over 11 months; wet: wet within 80 cm for over 6 months, but not wet within 40 cm for over 11 months.

are stored in vector format. After extracting the soil mapping units that fulfil the tree growth requirements (Table 2), the data were converted to raster format with a resolution of $1100\text{ m} \times 1100\text{ m}$.

All soil variables, except the “minimum soil water capacity”, stem directly from the European soil database. The “minimum soil water capacity” had to be defined through a pedotransfer function as described in Daroussin and King (1999).

2.4.3. Topographic variable

Topography (primarily altitude, exposition, and slope) or relief largely determines soil condition, erosion, micro- and meso-climate, and for example the suitability for agricultural use.

Data about topography were calculated from a digital elevation model (DEM) available at the United States Geological Survey (USGS, 2003). The dataset is called HYDRO 1K and has a resolution of 1 km^2 .

Altitude range expressed as metres above sea level was used to represent the topographic variable (Table 1).

2.5. Environmental risk areas

Environmental risk areas were defined as regions with a potential environmental problem due to agricultural land use. The environmental problems were classified as a high risk of soil erosion, nitrate vulnerable zones (NVZ) and low landscape diversity (uniform arable landscapes).

2.5.1. Soil erosion

The assessment of soil erosion was based on indicators and values produced by the PESERA project (Gobin and Govers, 2003). Erosion indicators were calculated using a model with a physical basis, applied to the regional scale. A preliminary map of the PESERA-project was reclassified to locate areas with more than 5 t/ha/year soil loss, which was defined as a high risk for soil erosion. The PESERA map was already in the correct resolution, but the projection had to be adapted to the projection of the PELCOM land cover map.

2.5.2. Nitrate vulnerable zones

Based on the Council Directive 91/676/EEC (European Commission, 2002), EU member states have identified “Nitrate vulnerable zones” where there is a high risk of nitrate leaching. The NVZ map is originally a vector map, developed at the European Environmental Agency and the Joint Research Centre (Ispra, Italy). It was transformed to a Grid-map with a resolution of 1100 m .

2.5.3. Landscape diversity

Uniform arable landscapes were selected from the two land-cover datasets CORINE (EEA, 2000) and PELCOM (Mücher, 2000). The CORINE land-cover map has a resolution of $250\text{ m} \times 250\text{ m}$, but is not available for all European countries. For the missing countries (Albania, Great Britain, Scandinavia, and Switzerland) the PELCOM map was used by re-sampling the PELCOM map to a pixel-size of 250 m . The re-sampled PELCOM map and the CORINE maps were merged. A moving window (focal majority command) was used to filter

out those areas in this new 250 m land cover map, where in a $5\text{ km} \times 5\text{ km}$ area more than 50% is covered by arable land. These areas were then reclassified as low diversity landscapes.

2.6. Target regions for silvoarable agroforestry

Target regions for silvoarable agroforestry for the five investigated trees were defined where productive tree growth on existing arable land was potentially possible and where at least one environmental problem existed (Fig. 1). The data model is illustrated in Fig. 2 and was processed using ArcInfo Macro Programming Language (AML). Once the criteria for the tree growth requirements were mapped and the maps for soil erosion, nitrate leaching, and landscape diversity were defined, all the maps were overlaid. GIS (Geographic Information System ArcInfo) was used to standardise, integrate, and query data on soil, climate, topography, and land cover. Once all datasets had been converted in raster format, the data were queried (using simple additions) to identify the areas where all the criteria were met. These areas were then defined as target regions.

2.7. Map-projection and resolution

Because the PELCOM land-cover map (Mücher, 2000) has a valid spatial reference on a European scale, all other datasets (Table 1) were transformed to the PELCOM coordinate system (Projection: Albers. Datum: WGS72).

With the exception of the climatic data, the DEM and the soil erosion map, all other datasets were transformed into Grid-maps with a resolution of $1100\text{ m} \times 1100\text{ m}$ in order to facilitate the overlay procedures.

3. Results and discussion

3.1. Research area and arable land use

The European countries which were analysed cover an area of $4.71 \times 10^6\text{ km}^2$. 34.5% of this area is used as arable land ($1.63 \times 10^6\text{ km}^2$).

3.2. Potential productive tree growth area

All together, the tree species investigated can potentially grow and produce economically viable wood yield on $906,887\text{ km}^2$. This corresponds to 55.8% of the arable land of Europe (Table 3).

3.2.1. Walnut (*Juglans* spp.)

Walnut trees are economically important because of their fruits and the decorative, valuable timber (Becquey, 1997). The definition of the potential productive growth area of walnut trees (*Juglans* hybrids) is based on the requirements of *Juglans regia*, which is found today in most of Europe except for northern and north-eastern regions (Norway, Finland, Poland). It grows well on fertile, deep, and well-drained soils (CABI, 2003).

Fig. 3 shows the potential productive growth area of walnut (*Juglans* hybrids). It covers an area of $242,961\text{ km}^2$. This is 14.9%

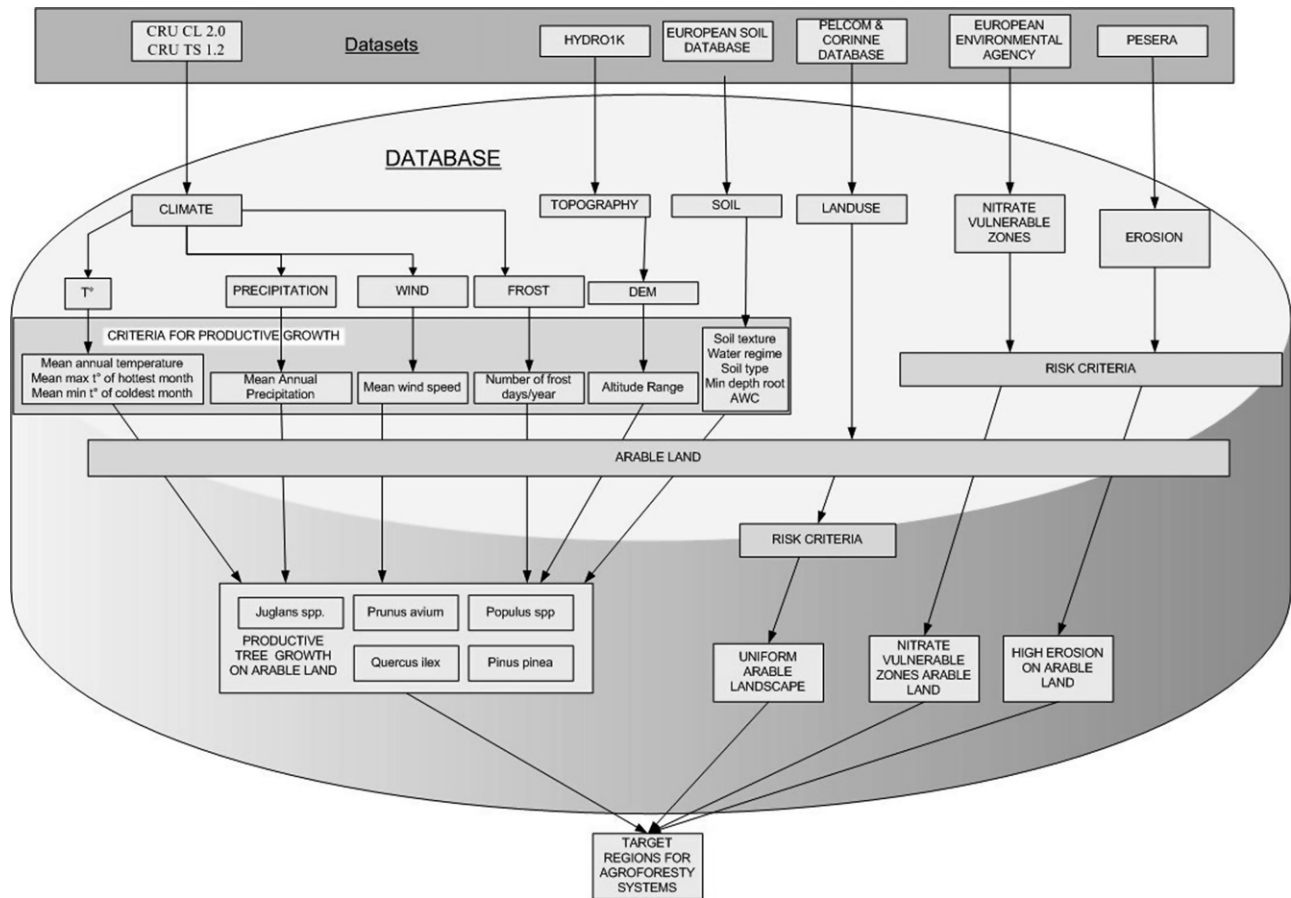


Fig. 2 – Database structure.

of the European arable land (Table 3). The main potential distribution is in flat regions of Germany, France, Italy, and the eastern part of Austria.

3.2.2. Wild cherry (*P. avium*)

P. avium is found on lowland plains and also on slopes and hills over most of Europe. It grows scattered along moist river valleys or in the edges of woods and in hedgerows (Ducci et

al., 1988). In general it needs a deep moist soil for good growth (Teissier du Cros, 1980; Savill, 1991). Current interest in wild cherry wood production in natural forests and in plantations in Europe is high, as it produces an attractive, patterned wood (Zimmermann, 1988; Schalk, 1990).

Fig. 4 shows the potential productive growth area of *P. avium*, which covers an area of 296,335 km² (18.2% of the European arable land, see Table 3). In some areas it has the same

Table 3 – Potential productive tree growth area and target regions for silvoarable agroforestry in Europe (for *Juglans* spp., *Prunus avium*, *Populus* spp., *Pinus pinea* and *Quercus ilex*)

Tree species	Potential productive tree growth area		Target regions	
	km ²	European arable land (%)	km ²	European arable land (%)
<i>Juglans</i> spp.	242,961	14.9	197,308	12.1
<i>Prunus avium</i>	296,335	18.2	222,604	13.7
<i>Populus</i> spp.	547,054	33.6	402,763	24.8
<i>Pinus pinea</i>	65,405	4.0	37,667	2.3
<i>Quercus ilex</i>	155,957	9.6	88,098	5.4
Area where at least one of the selected tree species can grow ^a	906,887	55.8	652,185	40.1

^a Because of overlaying, the area is smaller than the sum of the potential distribution area of the single tree species.

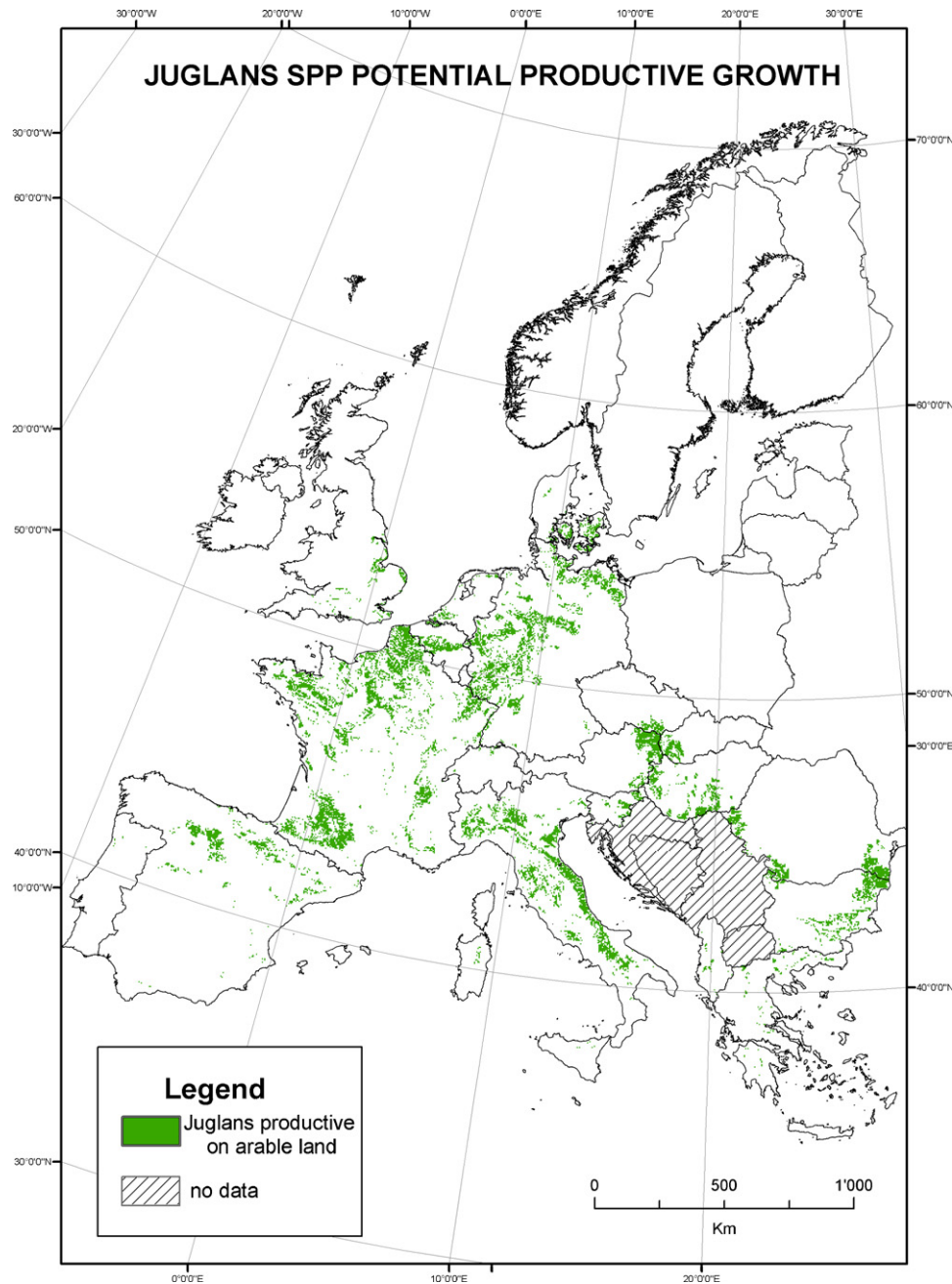


Fig. 3 – Walnut (*Juglans* spp.): potential productive tree growth area on arable land.

distribution as *Juglans regia*, but it stretches to colder regions and to regions with a more continental climate.

3.2.3. Poplar (*Populus* spp.)

The definition of target areas for poplar was based on the tree requirements of *Populus deltoides*, because it has been extensively used in poplar hybridization programmes throughout the world, producing fast-growing clones.

P. deltoides primarily grows on the moist alluvial soils along streams and on sandy or well-drained soils with a high water table which provides year-round moisture (Albertson and

Weaver, 1945; Dickmann and Stuart, 1983; Hupp, 1992; Kaul, 1995). The wood of poplar is commonly used for the manufacture of a number of products, including pallets, furniture, matches, and packing cases.

Fig. 5 shows the potential productive growth area of poplar hybrids. It covers 547,054 km² (33.6% of the European arable land, see Table 3). The potential productive tree growth area is very large. In reality it would be even larger, mainly in the Mediterranean region, as the resolution of the maps (1100 m × 1100 m) does not allow for the inclusion of smaller areas (e.g. along rivers and streams) into the analysis. More-

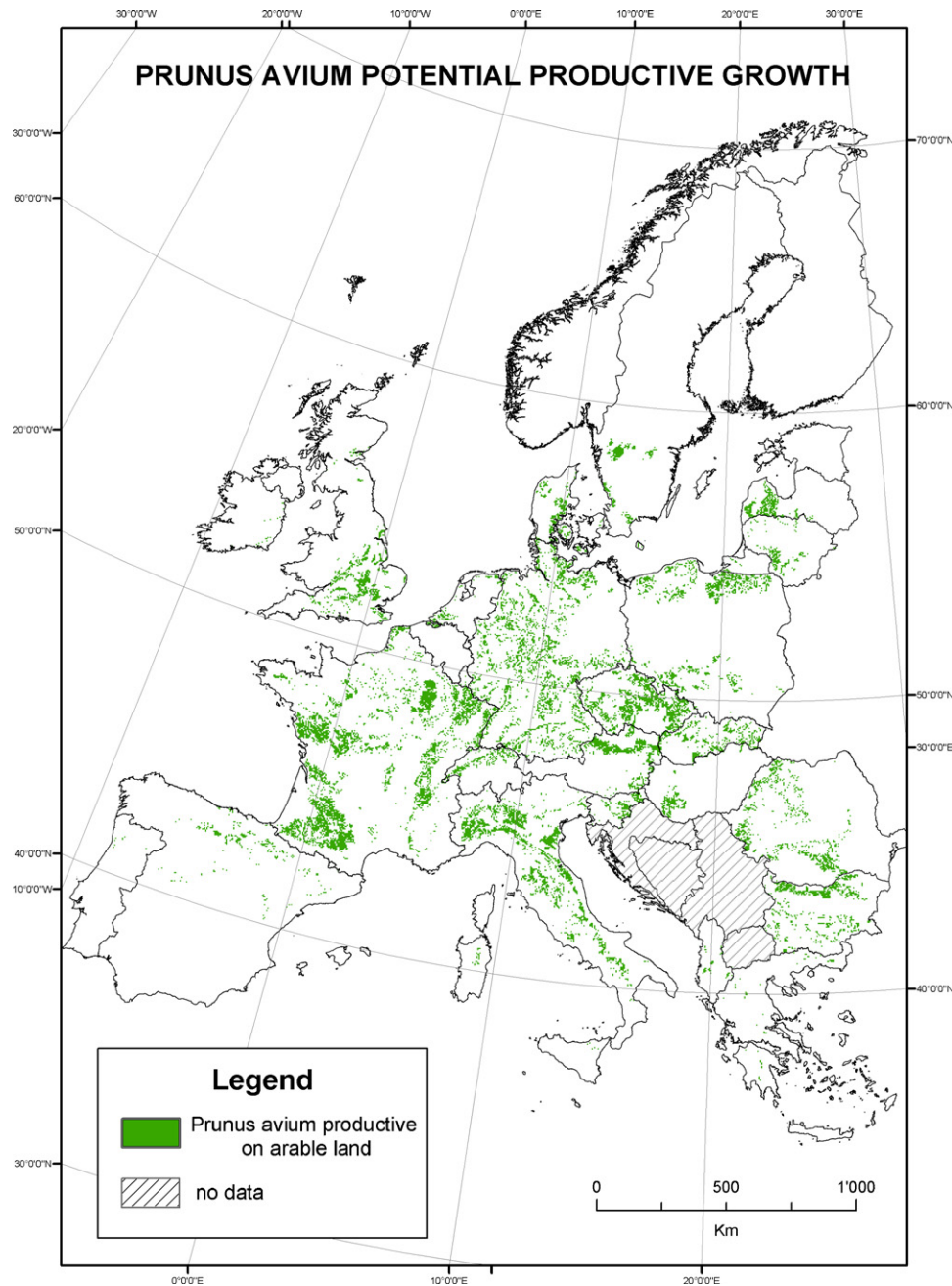


Fig. 4 – Wild cherry (*Prunus avium*): potential productive tree growth area on arable land.

over, irrigation would allow the extension of poplar planting into areas where rainfall was the limiting criterion.

3.2.4. Italian stone pine (*P. pinea*)

P. pinea has a distribution limited to the Mediterranean basin (Richardson and Rundel, 1998). It can grow on almost all soil types, including very poor soils, but it grows best on sandstone and sandy substrates (Barbéro et al., 1998). *P. pinea* is cultivated for different purposes such as fruits (seeds), solid wood, wood-fibre, and environmental protection (e.g. erosion control, drift sand control) (Le Maitre, 1998).

Fig. 6 shows the potential productive growth area of *P. pinea*, which covers an area of 65,405 km². This corresponds to 4.0% of the European arable land (Table 3).

3.2.5. Holm oak (*Q. ilex*)

Q. ilex is a typical Mediterranean tree. It is more important in the western part of the Mediterranean Basin than in the eastern part (Barbéro et al., 1992). *Q. ilex* can grow on a large range of soil types, from littoral sandy soils to granite soils, but it does not tolerate waterlogged conditions (CABI, 2003). The wood is used for firewood and to produce good-quality charcoal. The acorns of subsp.

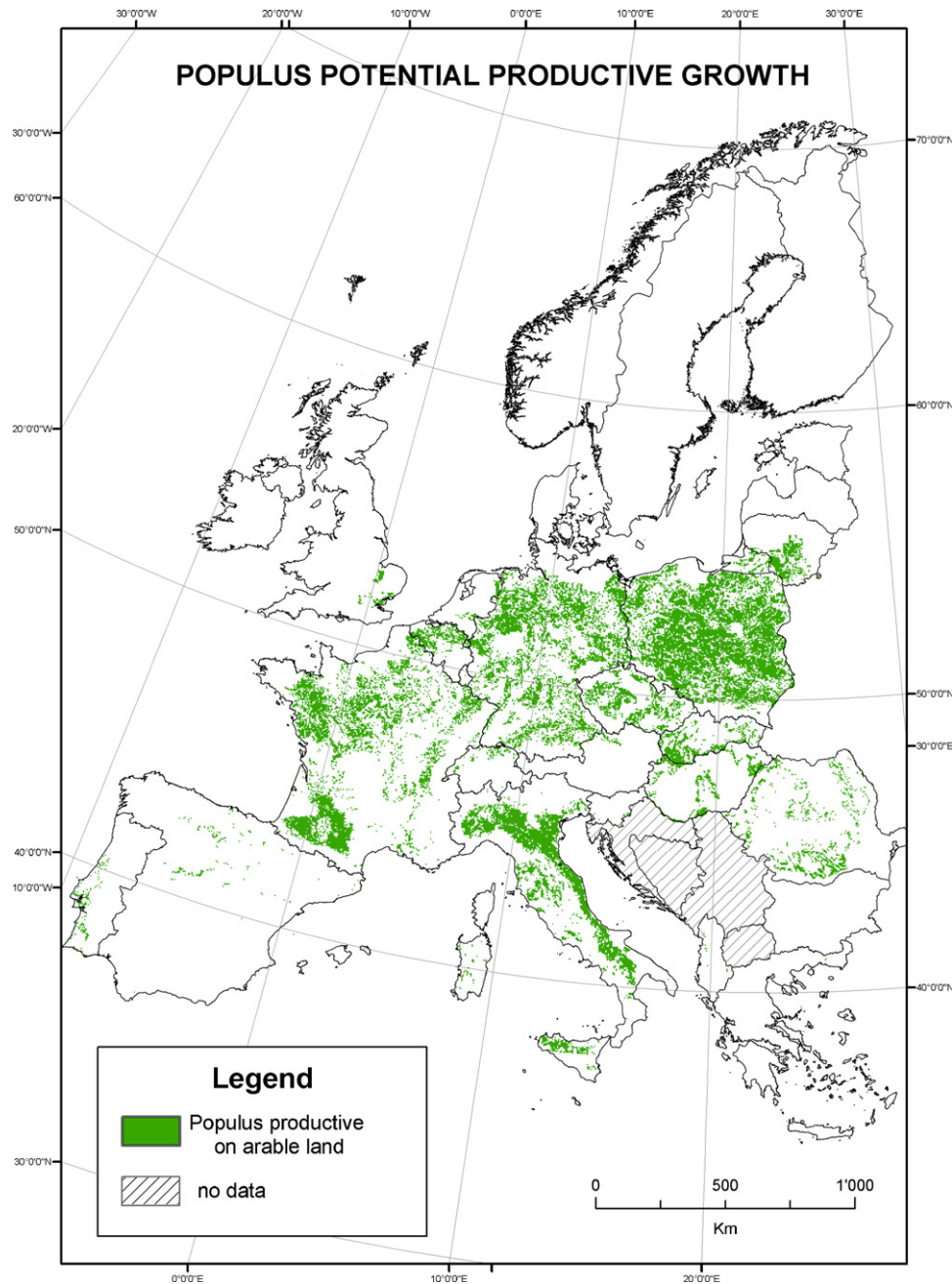


Fig. 5 – Poplar (*Populus* spp.): potential productive tree growth area on arable land.

rotundifolia are also collected and fed to animals (CABI, 2003).

Fig. 7 shows the potential productive growth area of *Q. ilex*. It covers 155,957 km² (9.6% of the European arable land, see Table 3).

3.3. Environmental risk on arable land

3.3.1. Soil erosion

The erosion map does not cover all selected countries (Section 2.2). The following countries were excluded because of insufficient data: Albania, Finland, Norway, Sweden, and Switzerland

(Fig. 8). The analysed countries cover 3.5×10^6 km², with an area of 1.53×10^6 km² arable land. 5.2% of the arable land has a high or very high risk of soil erosion with more than 5 t/ha/year (Table 4).

3.3.2. Nitrate vulnerable zones

Nitrate vulnerable zones (European Commission, 2002) have been defined for Austria, Belgium-Luxembourg, Denmark, Finland, France, Germany, Greece, Italy, The Netherlands, Portugal, Slovenia, Spain, Sweden, and The United Kingdom (Fig. 9). 984,383 km² in the analysed countries is arable land, of which 51.6% was assessed as NVZ with a high risk of nitrate leaching into the groundwater (Table 4).

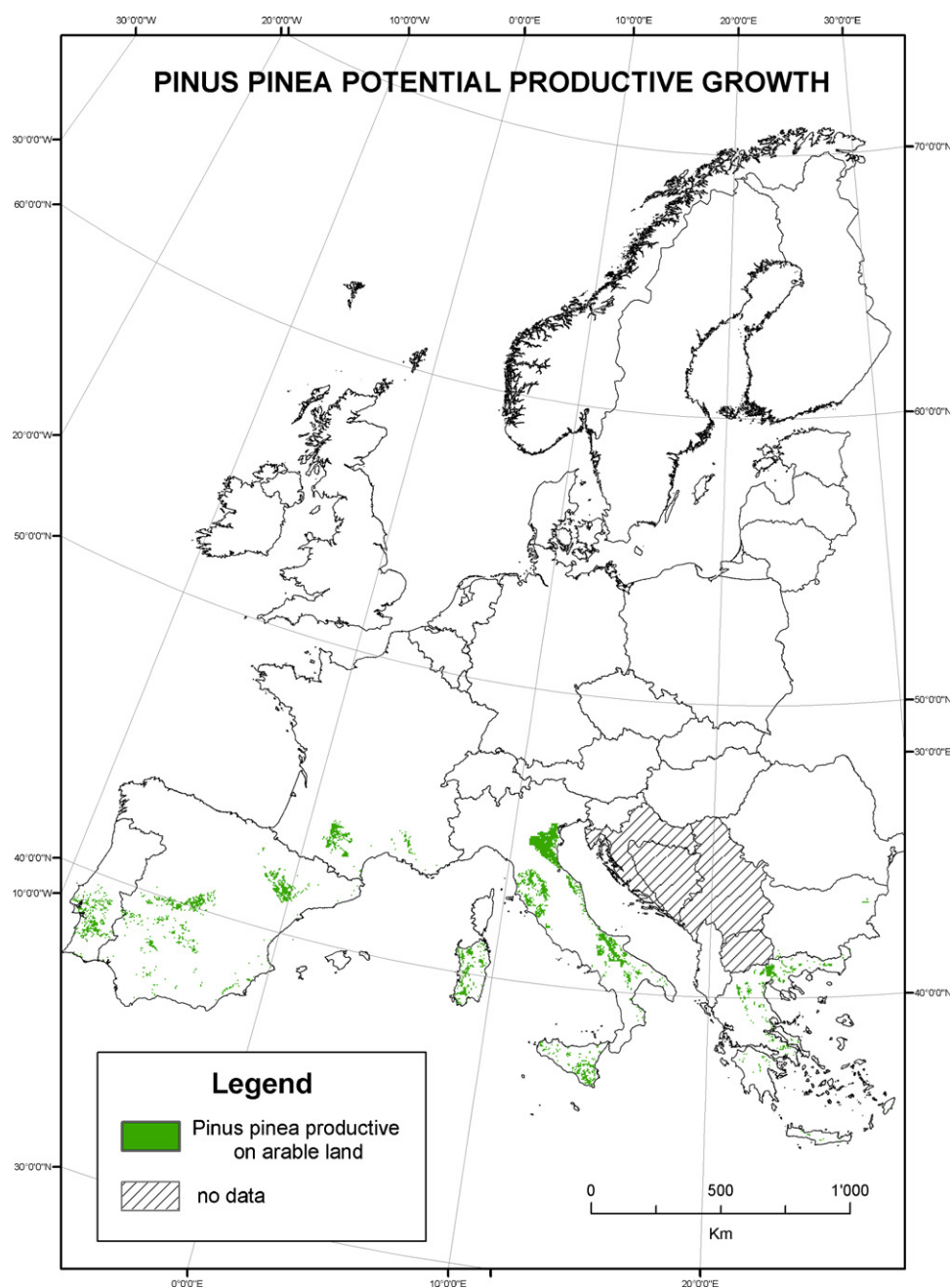


Fig. 6 – Italian stone pine (*Pinus pinea*): potential productive tree growth area on arable land.

Table 4 – Environmental risk areas on arable land in Europe (soil erosion, nitrate leaching and landscape diversity)

Environmental risk	Total arable land in selected countries ^a (km ²)	Risk area on arable land	
		km ²	Arable land ^a (%)
Soil erosion (>5 t/ha/year)	1,533,415	80,298	5.2
Nitrate leaching (NVZ)	984,383	508,019	51.6
Landscape diversity	1,626,389	1,077,521	66.3
Area where at least one environmental problem exists ^b	1,626,389 (Total European arable land)	1,304,083	80.2

^a See different countries taken into account in the maps Figs. 8–10.

^b Because of overlaying, the area is smaller than the sum of the potential distribution area of the single tree species.

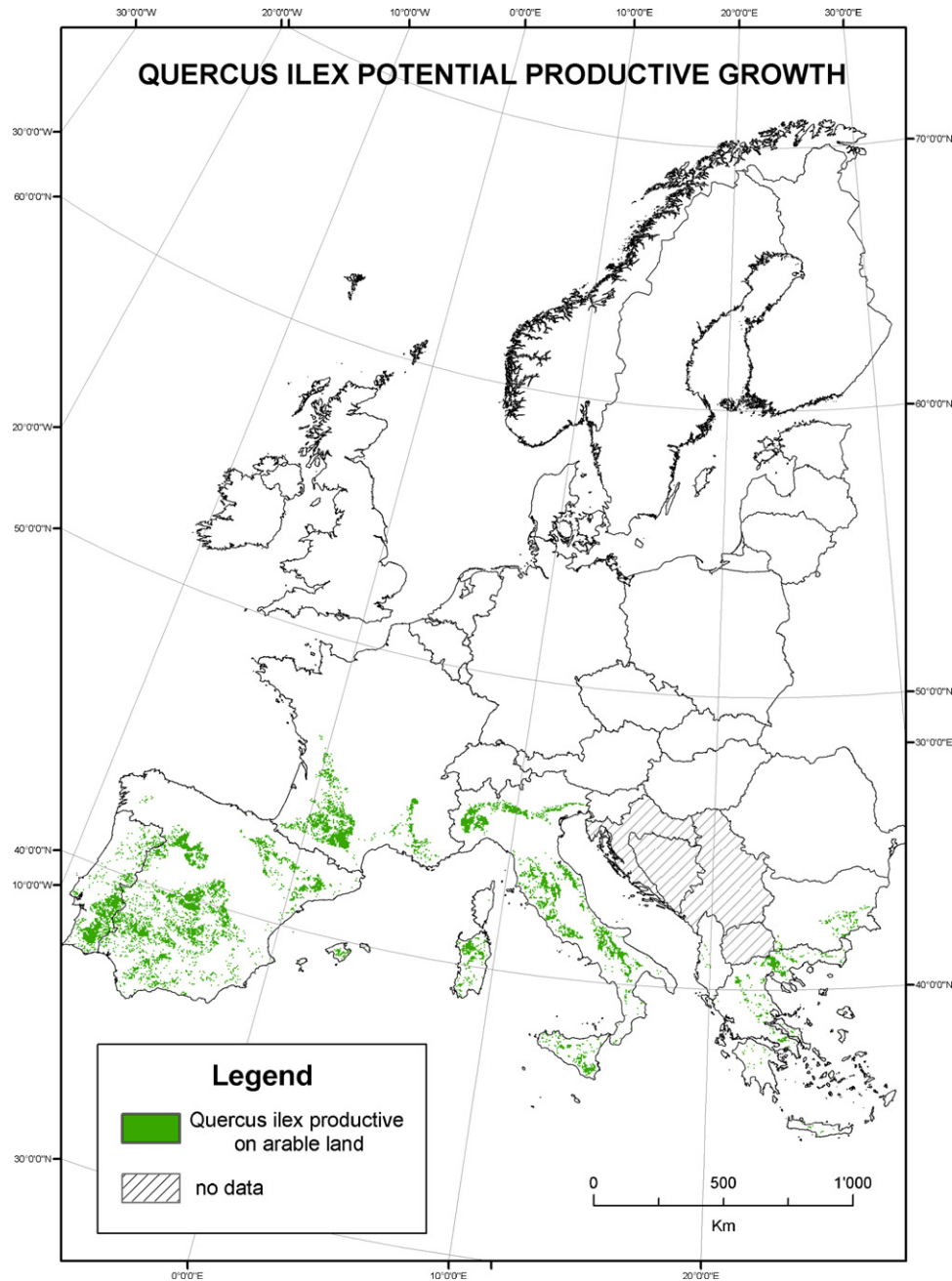


Fig. 7 – Holm oak (*Quercus ilex*): potential productive tree growth area on arable land.

3.3.3. Landscape diversity

All countries from the list in Section 2.2 were included. From the $1.63 \times 10^6 \text{ km}^2$ arable land, $1.08 \times 10^6 \text{ km}^2$ was assessed to have a low landscape diversity (Fig. 10). This represents 66.3% of the European arable land (Table 4).

3.4. Target regions for silvoarable agroforestry (with *Juglans spp.*, *P. avium*, *Populus spp.*, *P. pinea*, and *Q. ilex*)

Target regions resulted from overlaying the trees' potential growth areas with areas for which environmental problems were identified (Fig. 1). These target regions cover $652,185 \text{ km}^2$

(Table 3). This means that around 40.1% of the arable land in Europe was defined as a target region for at least one of the five tree species under investigation (Fig. 11). Of this area, 7% were classified as being in danger of erosion with an erosion rate of more than 5 t/ha/year . 34% of the target regions were categorised as nitrate vulnerable zones and 59% were found to have a uniform arable landscape.

If in the target regions new silvoarable agroforestry would be implemented, soil erosion could be reduced on 4% of the total area of European arable land. Similarly, nitrate leaching could be reduced on 18% of the arable land area in Europe and

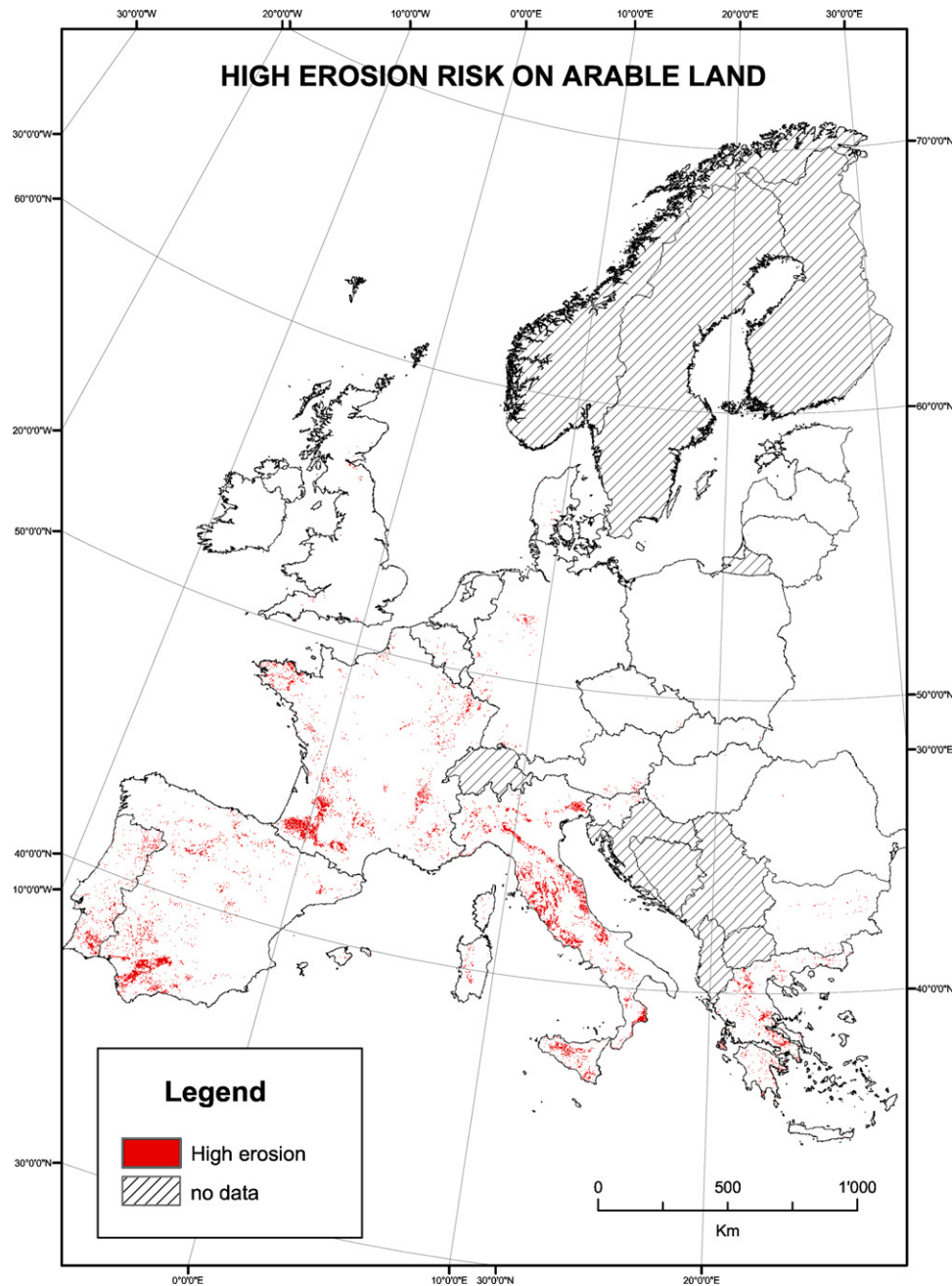


Fig. 8 – High soil erosion (>5 t/ha/year). Source: Gobin and Govers (2003).

32% of the arable European landscape could be diversified by trees.

Target regions cover large parts of central Europe. In many areas of Germany, Poland, and the Czech Republic, nitrogen leaching from arable agriculture could be reduced by integrating walnut, wild cherry, and/or poplar based agroforestry systems. The same is true of the arable landscapes along the river Danube in northern Austria and the agricultural areas near Bucharest. At the same time these rather monotonous agricultural landscapes would benefit from landscape diversification that the agroforestry system affords. Walnut, wild cherry, and poplar silvoarable agroforestry could also mitigate nitrate leaching in southern England, north-

ern (Brittany, Champagne) and central France (Paris basin, Loire region) and reduce soil erosion in the steeper zones of these regions. In southern France (Aquitaine, Pyrenees), walnut- and cherry-based agroforestry systems could have the same beneficial effects. In Italy, the Po valley would qualify for poplar agroforestry systems with the benefit of reducing nitrate leaching and increasing landscape diversity. This target region stretches all the way down the Italian peninsula on the eastern flank of the Apennine mountains. In this region, silvoarable agroforestry using walnut could be used mainly to reduce soil erosion.

In comparison to central Europe, fewer target regions were identified in the Mediterranean regions. This is partly due to

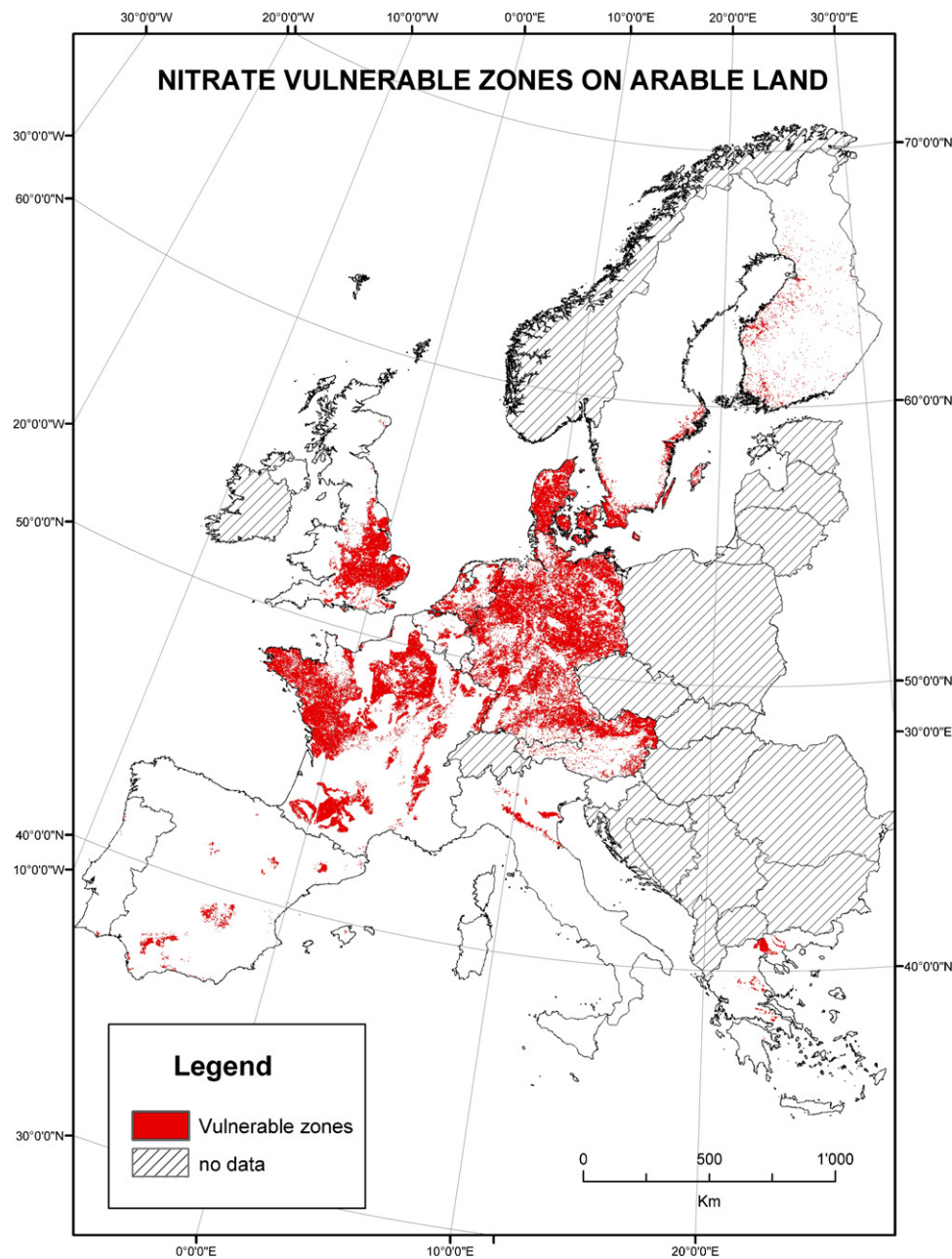


Fig. 9 – Nitrate vulnerable zones (NVZ). Source: European Commission (2002).

the fact that the maps (1 km²) are unable to locate small, narrow valley bottoms where e.g. Poplar could take advantage of the water table. Nevertheless, in southern Portugal, in Spain (Saragossa, Valladolid, Sevilla), in Sardinia, and in Sicily, the use of silvoarable agroforestry with poplar, oak, and pine could contribute to the reduction of soil erosion and also enhance landscape diversity.

3.5. Critical remarks

Both climate and soil conditions are important in determining the species' capability to grow successfully at a specific site. Climatic factors are particularly useful for indicating broad regions where specific species can grow. Within these

broad regions, soil characteristics constrain the suitability at a smaller scale. When interpreting the maps, it has to be kept in mind that they are based on coarse, European datasets with a resolution of 1 km². In reality, for specific sites, the local edaphic or microclimatic conditions can of course differ. The maps would probably look different if they were derived from national data of higher spatial and thematic resolutions. Therefore, we would recommend that national level analyses should be undertaken in order to increase the precision of the maps. Our analysis, however, has the merit of being applicable to the entire European continent.

The aim was to identify regions where a rapid and productive growth of the trees can be expected. As a consequence, the

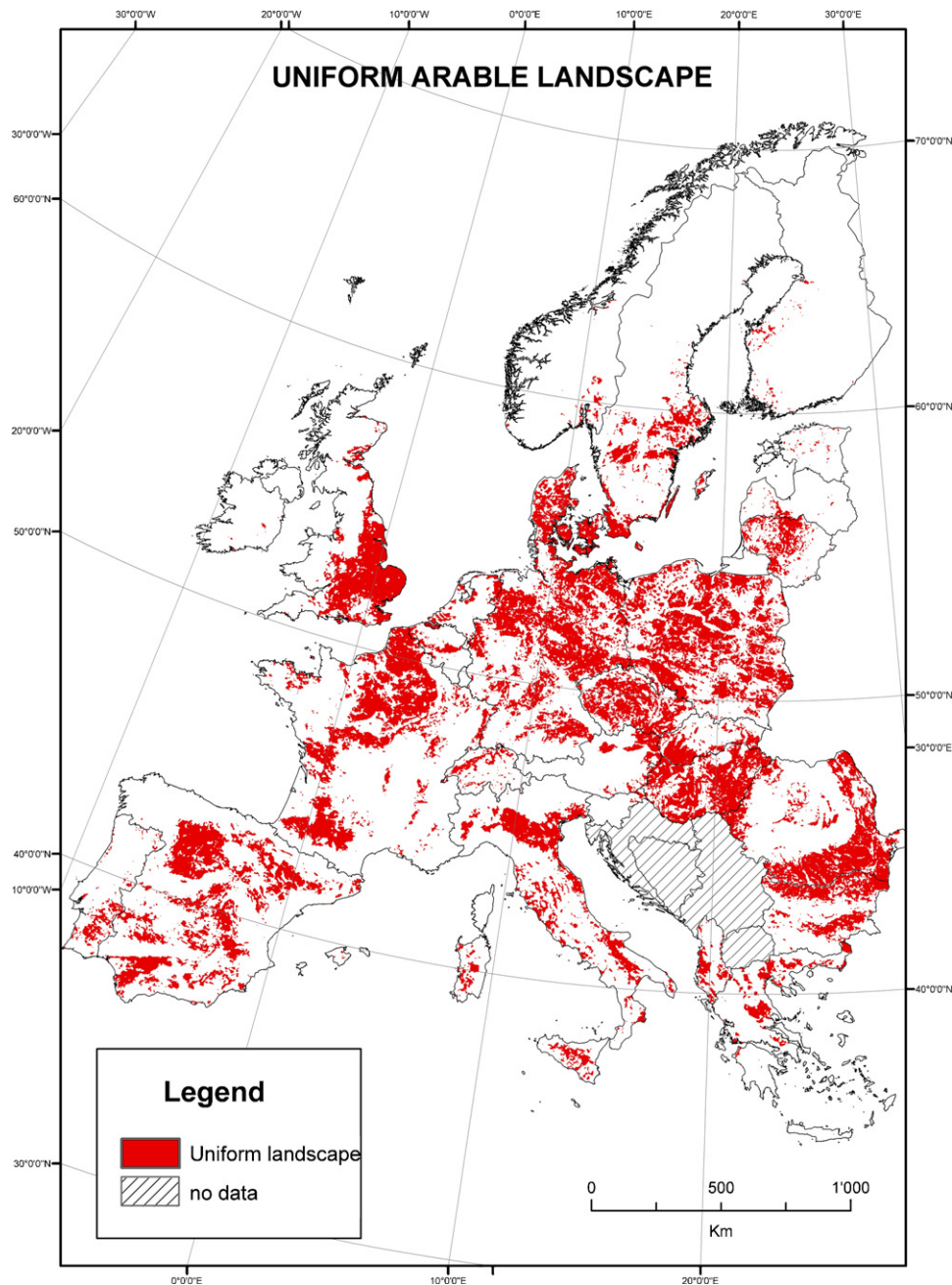


Fig. 10 – Low landscape diversity.

areas depicted on the maps (Figs. 3–7) are smaller than botanical maps (Jalas and Suominen, 1976; Clapham et al., 1987), which show the potential distribution of the tree species and which include marginal areas where the trees can grow, but will not be of economic interest in the context of a productive agroforestry system.

Potential growth areas are locations where the tree species are worth considering for planting trees in an agroforestry setting on arable land. But a tree species suited to a certain area may not be profitable due to other factors which were not included in this assessment, such as insects, diseases, weeds, limited availability of a suitable variety, and market access.

An important issue is the data availability. The description of environmental requirements of the five tree species investigated is based on a range of factors. The factors used here have been widely applied around the world to assist species selection (CABI, 2003). They have the advantage that the climatic requirements are based on simple monthly mean temperature and rainfall data, which are available for whole Europe. However, long-term average data does not reflect year-to-year variations, so problems may be experienced if trees are established for example in low rainfall areas during a drought period.

The assessment model created for this study has some limitation. The model relies on relatively few climate and soil

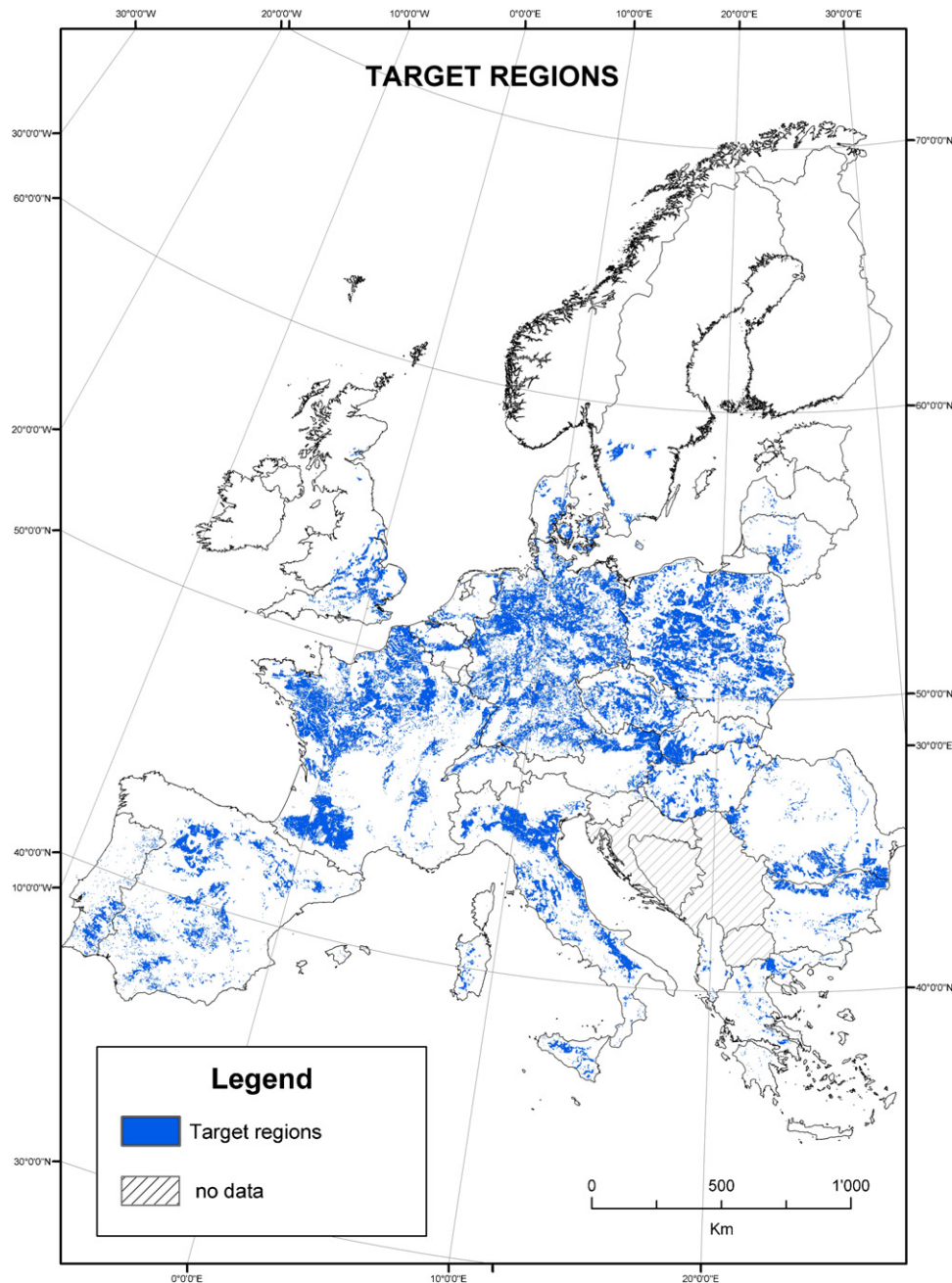


Fig. 11 – Target regions for silvoarable agroforestry in Europe for *Juglans* spp., *Prunus avium*, *Populus* spp., *Pinus pinea*, and *Quercus ilex*. Possible areas for agroforestry on arable land in Europe, where at least one of the five selected tree species can potentially grow, and where at least one environmental problem exists (soil erosion, nitrate leaching, landscape diversity).

requirements to create target regions maps. Requirements were selected because of their importance and availability. Additional requirements would strengthen the target regions maps but the necessary data are so far not available at the European level. An important aspect which would need improvement is the analysis of landscape diversity. Our coarse approach does not take into account differences between individual landscapes, local objectives for landscape development and landscape character. Such analysis would need to be conducted at higher spatial resolution for individual countries or regions.

4. Conclusions and outlook

The potential growth area of the five selected tree species covers large parts of temperate Europe (*Juglans* spp., *P. avium*, *Populus* spp.) and of Mediterranean Europe (*Populus* spp., *P. pinea*, *Q. ilex*). On about 906,887 km², at least one of the selected tree species can grow (Table 3). With 56% of the European arable land, the proportion of arable land where agroforestry could be implemented and productive tree growth expected is rather high.

The environmental impact of arable agriculture in Europe is considerable. 5% of the arable land has a high risk for soil erosion, 52% was classified as a nitrate vulnerable zone, and 66% was assessed to have low landscape diversity (note the different reference areas as described in Section 3.3). We estimated that on 1.30×10^6 km² (80% of the total European arable land) at least one of the three environmental risks existed (Table 4).

The target regions cover 652,185 km² which relates to approximately 40% of the arable land in Europe (Table 3). The study shows that productive growth of the investigated tree species can be expected throughout all climatic zones from north to south and from east to west. The development and introduction of new agroforestry techniques could contribute to the reduction of environmental risks on a considerable amount of the European arable land area.

If further tree species were taken into account, the target area for silvoarable agroforestry systems would increase. Notably coniferous species would need to be investigated for the boreal zone.

Our results indicate that silvoarable agroforestry systems deserve public support to at least the same extent as conventional agricultural and forestry production. Presently, as the European agricultural policy is revised and cross-compliance requirements are introduced, it is as yet unclear as to which degree member states will formulate regulations that favour the maintenance and even establishment of trees in agricultural landscapes. Our results demonstrate the potential of agroforestry at the continental scale. Further research should be conducted at higher spatial and thematic resolutions, taking into account the characteristics of individual landscapes in order to more precisely identify target regions for agroforestry. Specific recommendations could then be elaborated to eventually encourage the establishment of agroforestry in locations, where it is feasible and where it can contribute to resolve an environmental problem.

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REFERENCES

- Albertson, F.W., Weaver, J.E., 1945. Injury and death or recovery of trees in prairie climate. *Ecol. Monogr.* 15, 393–433.
- Anderson, L.S., Sinclair, F.L., 1993. Ecological interactions in agroforestry systems. *Agroforestry Abstr.* 54 (6), 57–91.
- Barbéro, M., Loisel, R., Quézel, P., 1992. Biogeography, ecology and history of Mediterranean *Quercus ilex* ecosystems. *Vegetatio* 99–100, 19–34.
- Barbéro, M., Loisel, R., Quézel, P., Richardson, D.M., Romane, F., 1998. Pines of the Mediterranean Basin. In: Richardson, D.M. (Ed.), *Ecology and Biogeography of Pinus*. Cambridge University Press, pp. 153–170.
- Becquey, J., 1997. *Les noyer à bois*. Les Guides du Sylviculteur. Institut Pour le Développement Forestier, Paris, 144 pp.
- Booth, T.H., Jones, P.G., 1998. Identifying climatically suitable areas for growing particular trees in Latin America. *Forest Ecol. Manage.* 108, 167–173.
- Booth, T.H., Jovanovic, T., 2002. Identifying climatically suitable areas for growing particular trees in Africa: an example using *Grevillea robusta*. *Agroforestry Syst.* 54 (1), 41–49.
- Booth, T.H., Jovanovic, T., New, M., 2002. A new climatic mapping program to assist species selection. *Forest Ecol. Manage.* 163, 111–117.
- Booth, T.H., Stein, J.A., Nix, H.A., Hutchinson, M.F., 1989. Mapping articular species: an example using Africa. *Forest Ecol. Manage.* 28, 19–31.
- Burgess, P.J., 1999. Effects of agroforestry on farm biodiversity in the UK. *Scot. Forestry* 53 (1), 24–27.
- Byington, E.K., 1990. Agroforestry in the Temperate Zone. In: MacDicken, K.G., Vergara, N.T. (Eds.), *Agroforestry, Classification and Management*. Wiley, New York, pp. 228–289.
- CABI, 2003. *Forestry Compendium*. C.A.B. International, Wallingford, UK. <http://www.cabi.org/compendia/fc/index.asp>.
- Carruthers, P., 1990. The prospects for agroforestry: an EC perspective. *Outlook Agric.* 19, 147–153.
- Clapham, A.R., Tutin, T.G., Moore, D.M., 1987. *Flora of the British Isles*, third ed. Cambridge University Press, 688 pp.
- Daroussin, J., King, D., 1999. Metadata: Pedotransfer rules database v.2.0 for environmental interpretations. The use of pedotransfer in soil hydrology research in Europe. In: *Workshop Proceedings, Orleans, France, 10–12 October 1996*, 12 pp.
- Dickmann, D.I., Stuart, K.W., 1983. *Poplar Culture in North America*. Michigan State University, Department of Forestry, East Lansing, MI, 168 pp.
- Ducci, F., Tocci, A., Veracini, A., 1988. Summary list of basic plant material of *Prunus avium* collected in N. and S. Italy. *Annali-Istituto Sperimentale per la Selvicoltura*, vol. 19. pp. 263–303.
- Dupraz, C., Newman, S.M., 1997. Temperate agroforestry in Europe. In: Gordon, A.M., Newman, S.M. (Eds.), *Temperate Agroforestry Systems*. C.A.B. International, Oxon, pp. 181–236.
- Dupraz, C., 1994. Prospects for easing land tenure conflicts with agroforestry in Mediterranean France: a research approach for intercropped timber orchards. *Agroforestry Syst.* 25 (3), 181–192.
- EEA, 2000. *CORINE land cover technical guide—addendum 2000*. Report No. 40. European Environmental Agency, Copenhagen, 105 pp.
- Ellis, E.A., Nair, P.K.R., Linehan, P.E., Beck, H.W., Blanche, C.A., 2000. A GIS-based database management application for agroforestry planning and tree selection. *Comput. Electron. Agric.* 27, 41–55.
- ESB, 2002. *European Soil Bureau*. European Soil Database version 2.
- ESRI, 2001. *ESRI Data and Maps*. Media Kit, Edition ArcInfo 8.1. CD-Rom.
- ESRI, 2002. *ArcGIS 8.2*. Environment System Research Incorporated, Redlands, California.
- Étienne, M., 1996. Western European silvopastoral systems. *Science Update*. Institut National de la Recherche Agronomique, Paris. INRA Editions, 276 pp.
- European Commission, 2002. *Implementation of Council Directive 91/676/EEC concerning the protection of waters against pollution caused by nitrates from agricultural sources*.

- Synthesis from Year 2000 Member States Reports. Luxembourg, 51 pp. <http://europa.eu.int/comm/environment/water/water-nitrates/91.676.eec.en.pdf>.
- Gobin, A., Govers, G., 2003. PESERA: Pan-European Soil Erosion Risk Assessment. Third Annual Report, 143 pp.
- Gordon, A.M., Newman, S.M. (Eds.), 1997. Temperate Agroforestry Systems. C.A.B. International, Oxon, 288 pp.
- Herzog, F., 1997. Konzeptionelle Überlegungen zu Agroforstwirtschaft als Landnutzungsalternative in Europa. Zeitschrift für Kulturtechnik und Landentwicklung 38 (1), 32–35.
- Herzog, F., 1998. Streuobst: a traditional agroforestry system as a model for agroforestry development in temperate Europe. Agroforestry Syst. 42 (1), 61–80.
- Herzog, F., 2000. The importance of perennial trees for the balance of northern European agricultural landscapes. Unasylva 200 (51), 42–47.
- Hijmans, R.J., Spooner, D.M., 2001. Geographic distribution of wild potato species. Am. J. Bot. 88 (11), 2101–2112.
- Hupp, C.R., 1992. Riparian vegetation recovery patterns following stream channelization: a geomorphic perspective. Ecology 73 (4), 1209–1226.
- Jalas, J., Suominen, J. (Eds.), 1976. Atlas Flora Europaea, Distribution of Vascular Plants in Europe, vol. 3. The Committee for Mapping the Flora of Europe and Societas Biologica Fennica Vanamo, Helsinki.
- Kaul, R.B., 1995. Reproductive structure and organogenesis in a cottonwood, *Populus deltoides* (Salicaceae). Int. J. Plant Sci. 156 (2), 172–180.
- Le Houérou, H.N., 1990. Agroforestry and sylvopastoralism to combat land degradation in the Mediterranean Basin: old approaches to new problems. Agric. Ecosyst. Environ. 33, 99–109.
- Lundgren, B., Raintree, J.B., 1982. Agroforestry. In: Conference of Directors of National Agroforestry Research Systems in Asia, Jakarta, 12 pp.
- Lundgren, B., 1982. What is agroforestry? Agroforestry Syst. 1 (1), 7–12.
- Le Maitre, D.C., 1998. Pines in cultivation: a global view. In: Richardson, D.M. (Ed.), Ecology and Biogeography of Pinus. Cambridge University Press, pp. 407–431.
- Metzger, M.J., Bunce, R.G.H., Jongman, R.H.G., Múcher, C.A., Watkins, J.W., 2005. A climatic stratification of the environment of Europe. Global Ecol. Biogeogr. 14 (6), 549–563.
- Mitchell, T.D., Carter, T.R., Jones, P.D., Hulme, M., New, M., 2004. A comprehensive set of high-resolution grids of monthly climate for Europe and the globe: the observed record (1901–2000) and 16 scenarios (2001–2100). Tyndall Centre Working Paper No. 55. Tyndall Centre for Climate Change Research, University of East Anglia, Norwich, UK, 30 pp. http://www.tyndall.ac.uk/publications/working_papers/wp55.pdf.
- Múcher, C.A. (Ed.), 2000. PELCOM project. Development of a consistent methodology to derive land cover information on a European scale from remote sensing for environmental modelling. Final Report. 236 pp.
- Múcher, C.A., Bunce, R.G.H., Jongman, R.H.G., Klijn, J.A., Koomen, A.J.M., Metzger, M.J., Wascher, D.M., 2003. Identification and Characterisation of Environments and Landscapes in Europe. Alterra-rapport 832, Alterra, Wageningen, 119 pp.
- Nair, P.K.R., 1993. An Introduction to Agroforestry. Kluwer, The Netherlands, 499 pp.
- New, M., Lister, D., Hulme, M., Makin, I., 2002. A high-resolution data set of surface climate over global land areas. Clim. Res. 21, 1–25.
- Palma, J., Herzog, F., Burgess, P.J., Graves, A.R., Keesman, K.J., van Keulen, H., Mayus, M., Reisner, Y., 2007. Methodological approach for the assessment of environmental effects of agroforestry at the landscape scale. Ecol. Eng. 29, 450–462.
- Pinto-Correia, T., 2000. Future development in Portuguese rural areas: how to manage agricultural support for landscape conservation? Landsc. Urban Plann. 50 (1–3), 95–106.
- Richardson, D.M., Rundel, P.W., 1998. Ecology and biogeography of Pinus: an introduction. In: Richardson, D.M. (Ed.), Ecology and Biogeography of Pinus. Cambridge University Press, pp. 3–45.
- Rowe, E.C., van Noordwijk, M., Suprayogo, D., Hairiah, K., Giller, K.E., Cadisch, G., 2001. Root distributions partially explain N-15 uptake patterns in Gliricidia and Peltophorum hedgerow intercropping systems. Plant Soil 235 (2), 167–179.
- Sanchez, P.A., 1995. Science in agroforestry. Agroforestry Syst. 30 (1–2), 5–55.
- Savill, P.S., 1991. The Silviculture of Trees Used in British Forestry. C.A.B. International, Wallingford, UK, 143 pp.
- Schalk, P.H., 1990. Prunus in forest and landscape in the Netherlands. Nederlands Bosbouw tijdschrift 62 (5), 144–151.
- Shakesby, R.A., Coelho, C.O.A., Schnabel, S., Keizer, J.J., Clarke, M.A., Contador, J.F.L., Walsh, R.P.D., Ferreira, A.J.D., Doerr, S.H., 2002. A ranking methodology for assessing relative erosion risk and its application to dehesas and montados in Spain and Portugal. Land Degrad. Dev. 13 (2), 129–140.
- Sinclair, F.L., 1999. A general classification of agroforestry practice. Agroforestry Syst. 46 (2), 161–180.
- Somarrriba, E., 1992. Revisiting the past: an essay on agroforestry definition. Agroforestry Syst. 19, 233–240.
- Teissier du Cros, E., 1980. Improvement of broadleaved species: the present state of the art in West Germany and France. Rev. Forestière Française 32 (2), 149–166.
- Telleria, J., Santos, T., 1995. Effects of forest fragmentation on a guild of wintering passerines: the role of habitat selection. Biol. Conserv. 71, 61–67.
- USGS, 2003. U.S. Geological Survey, HYDRO 1K Europe, Elevation Derivative Database. <http://edcdaac.usgs.gov/gtopo30/hydro/europe.asp>.
- van Noordwijk, M., Lawson, G., Soumare, A., Groot, F., Hairiah, K., 1996. Root distribution of trees and crops: competition and/or complementary. In: Huxley, P. (Ed.), Tree-Crop Interaction: A Physiological Approach. Cambridge University Press, pp. 319–364.
- Young, A., 1997. Agroforestry for Soil Management. C.A.B. International and ICRAF, Wallingford, UK, 306 pp.
- Zimmermann, H., 1988. The importance and management of wild cherry. Allgemeine Forstzeitschrift 20, 538–540.